

Essential Insights: Week 1.1



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Why Distinguish AI Paradigms?

The Central Challenge in Organisations

- AI is often treated as a single capability, leading to confusion and inconsistent business value
- Discussions at board level are common, but without clear distinctions, initiatives fall short

Benefits of Understanding Differences

- Enables effective strategic decisions and responsible AI adoption
- Shifts focus from experimentation to embedding AI thoughtfully in workflows

Weekly Overview

- Explores classical ML (prediction), generative AI (creation), and agentic AI (action) as complementary tools

Overview of the Three AI Paradigms

Classical Machine Learning (Predictive AI)

- Analyses historical data to identify patterns and forecast future outcomes
- Provides probability scores or estimates to support human decisions, without taking action itself

Generative AI

- Shifts from prediction to creating new content like text, images, code, or drafts
- Boosts productivity in knowledge work, but requires human oversight for accuracy and bias

Agentic AI

- Advances to planning and executing actions towards specific goals, coordinating multiple steps
- Operates within predefined boundaries, with governance for oversight and escalation

Complementary Integration

- Combine paradigms: Predict outcomes, generate content, then act autonomously for maximum leverage

Classical ML: Focus on Prediction

Core Mechanism

- Learns from historical data to estimate likely future events, outputting forecasts or risk scores
- Reduces uncertainty in decision-making, especially at large scales

Key Limitations

- Relies on past patterns; performs poorly when conditions change dramatically
- Does not generate new ideas, make decisions, or act independently

Strategic Value

- Embedded in operations for years, driving financial and efficiency gains through better-informed judgements

Real-World Use Cases of Classical ML

Demand Forecasting in Retail

- Uses sales history, promotions, events, and weather data to predict product needs across stores
- Improves inventory management, reducing stock-outs and excess holdings (e.g., Walmart's scalable models)

Credit Risk Assessment in Banking

- Analyses repayment behaviours and economic indicators to calculate default probabilities
- Enables precise pricing and loss reduction in large loan portfolios (e.g., JPMorgan Chase)

Fraud Detection in Payments

- Assesses transactions in real-time based on historical patterns, flagging or escalating risks
- Handles high volumes automatically, leaving complex cases for human review

Common Thread

- Predictions inform actions, but responsibility stays with humans and rules

Strengths of Classical ML

Ideal Environments

- Thrives in stable, data-rich settings with structured information and clear objectives
- Delivers consistent, measurable improvements in efficiency and accuracy

Case Study: Logistics Optimisation

- UPS's ORION system predicts optimal delivery routes using historical routes and traffic data
- Achieves significant savings in fuel and costs by minimising daily miles driven

Broader Benefits

- Scalable for repeatable processes, often invisible to end-users but impactful on margins
- Enhances decision quality without replacing human oversight

Limitations of Classical ML

Dependence on Historical Data

- Assumes past patterns predict the future; fails during structural shifts like pandemics
- COVID-19 example: Demand models became unreliable as consumer behaviours changed abruptly

Handling Complexity

- Struggles with unstructured qualitative data, such as news or policy changes
- Requires retraining and human intervention in volatile periods

Practical Insights for Leaders

- Deploy in predictable domains; avoid over-reliance to prevent misplaced confidence
- Complements other paradigms for addressing novelty and execution

Generative AI: The Shift to Creation

Fundamental Change from Prediction

- Moves AI from estimating outcomes to producing usable content based on prompts and context
- Accelerates workflows by generating drafts, ideas, or visuals directly

Integration with Predictive AI

- Builds on forecasts: E.g., predictive models identify audience preferences, generative tools create tailored content

Illustrative Example: Marketing Innovation

- Coca-Cola's 2023 campaign used generative AI for over 100,000 customised digital artworks
- Enables rapid iteration and scale, but demands human review for brand alignment and nuance

Early Business Impact of Gen AI

Drivers of Rapid Adoption

- User-friendly interfaces allow natural language inputs for instant outputs
- Reduces time and effort in content creation, broadening access beyond specialists

Applications Across Functions

- Marketing: Generates copy and visuals with safeguards (e.g., Canva, Adobe Firefly)
- E-commerce: Automates product descriptions and emails (e.g., Shopify for small teams)
- Consulting: Supports research synthesis and reporting (e.g., McKinsey for efficiency gains)
- HR: Drafts job postings and policy summaries, freeing time for strategic work

Strategic Outcomes

- Enhances experimentation, throughput, and participation; viewed as augmentation requiring governance

Key Changes with Generative AI

From Insight to Execution

- Predictive AI focuses on 'what might happen' to inform decisions
- Generative AI addresses 'what to create now' to support direct action in workflows

Complementary Use in Practice

- Marketing: Predict engagement, then generate campaign variations
- Sales: Forecast deal likelihood, draft personalised proposals
- Customer Support: Prioritise tickets, create responses and explanations

Overall Transformation

- Delivers value at execution stage, promoting productivity but necessitating quality controls
- Encourages combining paradigms for seamless insight-to-action flows

Agentic AI: Advancing to Autonomous Action

Core Capabilities

- Goes beyond prediction and creation to plan, coordinate, and execute multi-step processes
- Achieves goals like updating systems or notifying teams, within strict boundaries

Synergy with Other Paradigms

- Leverages predictive forecasts and generative outputs for informed, effective actions
- Example: Identify risks (predictive), draft reports (generative), then implement changes (agentic)

Governance Priorities

- Essential for increased autonomy: Define rules, ensure oversight, and include escalation paths
- Balances innovation with responsibility in high-stakes environments

Leadership Mindset

Recap of Paradigms

- Classical ML: Reduces uncertainty through data-driven predictions
- Generative AI: Accelerates creation and embeds in daily workflows
- Agentic AI: Enables coordinated actions for enhanced execution

Strategic Integration

- Paradigms are complementary; select based on task needs for greatest leverage
- Combine for holistic value: From decision support to autonomous operations

Mindset Shift for Leaders

- Ask: Where does prediction fit? How can creation speed work? When to enable action?
- Focus on intentional application, clear objectives, and robust governance for sustainable impact